

Literature Review on Home Automation System

Prof. Agalya P¹, Sharan N¹

Department of Electronics and Communication Engineering

Sapthagiri College of Engineering Bangalore-57, India

Affiliated to Visvesvaraya Technological University, Belagavi, KA, India

Abstract: *One of the topics which is gaining popularity is Home Automation System because of its numerous advantages. Home automation refers to the monitoring and controlling of home appliances remotely. With the never-ending growth of the Internet and its applications, there is much potential and scope for remote access and control and monitoring of such network enabled appliances. This paper deals with discussion of different intelligent home automation systems and technologies from a various features standpoint. The effort targeted on the home automation concept of where the controlling and monitoring operations are expediting through smart devices. Wide-ranging home automation systems and technologies considered in review with central controller based (Arduino or Microcontroller), cloud-based, Bluetooth-based, SMS based, ZigBee based, mobile-based, RF Module based, web based and the Internet with performance.*

Keywords: *Home Automation, Intelligence, Microcontroller, Sensor System, User-friendly Interface.*

I INTRODUCTION

Automation performs an increasingly vital role in daily experience and global economy. Engineers strive to combine automated devices with mathematical and organizational tools to create complex systems for a rapidly expanding range of applications and human activities.

A **Home Automation System** refers to the use of technology to automatically control household appliances and systems such as lighting, fans, air conditioners, security

cameras, door locks, entertainment systems, and more. It integrates hardware (sensors, controllers, smart devices) and software (mobile apps, voice assistants, cloud services) to provide comfort, safety, efficiency, and convenience in daily life.

The system allows homeowners to monitor and manage their home environment either **locally (within the house)** or **remotely (via smartphones, tablets, or computers)** using internet connectivity. With advancements in the Internet of Things (IoT), Artificial Intelligence (AI), and wireless communication, home automation has become more affordable and user-friendly.

Challenges of Home automation systems

Home automation systems suffer from four main challenges; these are poor manageability, inflexibility, difficulty in achieving security, and high cost of ownership. The main objectives of this research is to design and implement a home automation system using IoT that is capable of controlling and automating most of the house appliances through an easy manageable web interface. The proposed system has a great flexibility by using Wi-Fi technology to interconnect its distributed sensors to home automation server. This will decrease the deployment cost and will increase the ability of upgrading, and system reconfiguration.

II. LITERATURE SURVEY

1. Bluetooth based home automation system using cell phones:

In Bluetooth based home automation system the home appliances are connected to the Arduino BT board at input output ports using

relay. The program of Arduino BT board is based on high level interactive C language of microcontrollers; the connection is made via Bluetooth. The password protection is provided so only authorized user is allowed to access the appliances. The Bluetooth connection is established between Arduino BT board and phone for wireless communication. In this system the python script is used and it can install on any of the Symbian OS environment, it is portable. One circuit is designed and implemented for receiving the feedback from the phone, which indicate the status of the device.

2. Zigbee based home automation system using cell phones:

To monitor and control the home appliances the system is designed and implemented using Zigbee. The device performance is record and store by network coordinators. For this the Wi-Fi network is used, which uses the four-switch port standard wireless ADSL modern router. The network SSID and security Wi-Fi parameter are preconfigured. The message for security purpose first process by the virtual home algorithm and when it is declared safe it is re-encrypted and forward to the real network device of the home. Over Zigbee network, Zigbee controller sent messages to the end. The safety and security of all messages that are received by the virtual home algorithm. To reduce the expense of the system and the intrusiveness of respective installation of the system Zigbee communication is helpful.

3. Wi-Fi based home automation system using cell phones:

Wi-Fi based home automation system mainly consist three modules, the server, the hardware interface module, and the software package. The figure shows the system model layout. Wi-Fi technology is used by server, and hardware Interface module to communicate with each other. The same technology uses to login to the server web-based application. The server is connected to the internet, so remote users can access server web-based application through the internet using compatible web browser.

Software of the latest home automation system is split to server application software, and Microcontroller (Arduino) firmware. The Arduino software, built using C language, using IDE comes with the microcontroller itself. Arduino software is culpable for gathering events from connected sensors, then applies action to actuators and preprogramed in the server. Another job is to report the and record the history in the server DB. The server application software package for the proposed home automation system, is a web-based application built using asp.net. The server application software can be accessed from internal network or from internet if the server has real IP on the internet using any internet navigator supports asp.net technology. Server application software is culpable of, maintain the whole home automation system, setup, configuration. Server use database to keep log of home automation system components, we choose to use XML files to save system log.



Fig.The proposed home automation system layout

4. Home automation using RF module:

The main purpose of a Home Automation System using an RF module is to control household appliances with a remote. As technology advances, homes are becoming smarter, moving away from traditional wall switches to centralized control systems with RF-based switches.

In many houses, switches are spread out in different rooms, making it inconvenient for users to move around to operate them. This becomes even more challenging for elderly or physically disabled people. To solve this problem, RF technology offers a simple and efficient solution.

In this system, a remote control with an RF transmitter is used. It sends ON/OFF

commands to a receiver connected to the microcontroller, which then operates the appliances. By pressing buttons on the RF remote, users can easily turn devices ON or OFF wirelessly from anywhere within the coverage area.

5. Cloud Based home automation system:

Home Automation using cloud based system focuses on design and implementation of home gateway to collect data about data from home appliances and then send to the cloud-based data server to get store on Hadoop Distributed File System, it is process using MapReduce and use to implement a monitoring tasks to Remote user Presently home Automation System is persistently developing its resilience by assimilating the current characteristics which gratify the rising interest of the people. This paper presents the design and development of home automation system that use the cloud computing as service. The current system consists of three important units: the first part is cloud server, handle and controls the data and information of client and users and the status of devices The hardware interface module is the second part which implement the relevant connection to the actuators and sensing devices which give the physical service. Last part is Home Server, which construct the hardware device and gives the user interface. This paper focus to build the web services using cloud which is need for security and storage and availability of the data. The current system is cost efficient, reliable and comfortable which also gives a secured home automation system for entire family.

The system is made up of various client modules for various platforms.

1. Cloud server Cloud Server is a central server aims on implementing services to the other sub

modules. Central server serves as the data respiratory system and brain It implements three connections to the three sub modules vizhome system, web configuration tool and mobile. The server evaluates the data it takes from the house, send current status to the mobile device and vice versa. A database is managing by the server and it is status gets updated as per the changes done at home end.

2. Embedded Program for Hardware Circuit Microcontroller, and

3. Internet Client for any desktop or mobile phones.

6. Wireless Home Automation system using IoT:

This system uses mobiles or computers to control basic home control and function automatically through internet from anywhere around the world globally, an automated home is sometimes called a smart home. It is meant to save the electric power and human energy. The proposed system is a distributed home automation system, consists of server i.e. Wi-Fi module, sensors. Server controls and monitors the various sensors, and can be easily configured to handle more hardware interface module (sensors). The Arduino board, with built in Wi-Fi module acts as web server. Automation System can be accessed from the web browser of any local PC using server IP, or remotely from any PC or mobile handheld device connected to the internet with appropriate web browser through server real IP (internet IP). Wi-Fi technology is selected to be the network infrastructure that connects server and the sensors. Wi-Fi is chosen to improve system security (by using secure Wi-Fi connection), and to increase system mobility and scalability.

III. COMPARISON

Sl.No	System	Communication Interface	Controller	User Interface	Applications	Merits
-------	--------	-------------------------	------------	----------------	--------------	--------

1	Bluetooth based home automation system using cell phones	Bluetooth Module	Arduino	Python supported mobile	controlling	Secured and Low cost
2	Zigbee based home automation system using cell phones	Zigbee Network	Microcontroller	Smartphone App	entrance control management , monitoring the power consumption , temperature and humidity	Convenience, safety, and Power saving
3	Wi-Fi based home automation system using cell phones	Wi-Fi	Arduino, Sensors & Actuators	Wi-Fi module, Phone Application	Appliances Control, Environmental Monitoring, Security.	Flexibility & Scalability, Cost-Effectiveness.
4	Home automation using RF module	RF modules (ASK)	Microcontrollers	Remote Control, Mobile Application with IoT integration	Lighting control, Security Systems, Energy Management	Accessibility, Energy Efficiency, Ease of Installation.
5	Cloud Based home automation system	Wireless Technologies	Cloud based data server	Voice Control, Mobile app	Monitoring and Controlling Home Appliances	Convenience, Energy saving, Comfort, Cost Reduction.
6	Wireless Home Automation system using IoT	Wi-Fi	Arduino	Web Application and android App	Temperature and motion detection, monitoring and controlling appliances	Low cost, Secure, remotely controlled
7	Wireless Sensors Based with mobile Technology	cloud-based data server	PCB circuits	Mobile Application	monitor the home conditions and power consumption of appliance	Low power consumption and system cost efficiency.

IV CONCLUSION

Home automation systems have rapidly evolved with different wireless technologies such as **Bluetooth, Zigbee, Wi-Fi, RF modules, Cloud platforms, IoT, and sensor-based mobile technologies**. Each approach offers its own advantages — Bluetooth

provides low-cost short-range control, Zigbee ensures low power consumption and reliable networking, Wi-Fi offers high-speed connectivity, RF modules enable simple remote operation, Cloud systems provide remote monitoring and scalability, IoT integrates intelligent devices for smart living, and wireless sensors enhance automation and adaptability.

Together, these systems aim to improve **convenience, comfort, security, and energy efficiency** in modern homes. With continuous advancements in wireless communication and smart devices, home automation is becoming more accessible, user-friendly, and essential, ultimately transforming traditional houses into **smart homes of the future**.

REFERENCES

- [1] Bluetooth Based Home Automation Using Arduino-Veer Uday Shantaram, More Tushar Sakharan, Mohite Vishal Ramesh, Suryawanshi Rushikesh Bajirao (2022)
- [2] *Home automation and control using the ZigBee protocol* — Voicu Babiciu Published 2023/2022; Student Thinkers and Advanced Research
- [3] M. A. Omran, “*The design and fulfillment of a Smart Home (SH) material ...*”, 2022 — Design and prototype of a Wi-Fi based home automation controlled from an Android smartphone (ESP/Wi-Fi module + mobile app). ScienceDirect
- [4] *IoT BASED HOME AUTOMATION SYSTEM WITH WI-FI (ESP8266)*, IJCRT, 2024 — Project paper describing an ESP8266 Wi-Fi implementation that lets users control appliances via smartphone over the local network / internet. Good for implementation details (schematics, code snippets)
- [5] Voice Controlled Home Automation with Cloud-Based Environment Monitoring System-Md. Mutasim B.A.N. Akanda, M. Shiam Prodhan, Sabrina Sarwar, Arunangshu Mojumder Raatul & Bijan Paul — 2023
- [6] Energy Management System using RF module 2025; ICREGCSD 2025.
- [7] Home Security System Using Wireless Sensors Network — Wireless Personal Communications, 2022**